YAHYA HALLANI

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EDUCATION

McGill University May 2026 - May 2030 (Expected)
Bachelor of Engineering, Software Engineering (Co-op) / Montreal, QC

Collège André-Grasset Sep 2024 - May 2026
Diploma of College Studies (DCS), Mathematics, Science & Informatics / Montreal, QC
- Lead Programmer, Grasset Robotics Team - Canadian Robotics Competition (CRC)

AWARDS & COMPETITIONS

Perfect Score - 75/75, Canadian Computing Competition, University of Waterloo 2026

1st Place - DIRO Hackathon, Université de Montréal (1st of 18 teams from 9 CEGEPs; $2,500 team scholarship) 2026

1st Place - Intercollegiate Video Game Creation Competition (Atlantis, 2D game built in Godot 4) 2026

1st Place - Canadian Robotics Competition, Programming (lead programmer); 2nd Place the next year 2025, 2026

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Rust, GDScript, SQL, JavaScript, Java
Frameworks & Libraries: FastAPI, Flask, PySide6/Qt, NumPy, pytest, Playwright, Godot 4, React
Tools & Platforms: Git, GitHub Actions, SQLite, Linux, Bash, VS Code, Docker
Spoken Languages: French (native), English (fluent), Arabic (fluent), Spanish (intermediate)

EXPERIENCE

Mercor Sep 2026 - Present
Coding Agent Evaluator (Contractor) / Remote | Contract: Exceptional Software Engineers (Coding Agent Experience)
- Earned a promotion and a **100% approval rate** on submitted evaluations for a leading AI lab by comparing 2 coding agents per task on enterprise-scale public repositories, delivering 2 detailed reports (4 model outputs) per day during sprints.
- Caught recurring **verification failures, hallucinations and false success claims** by scoring agents on instruction following, code quality, thoroughness and speed, documenting each issue with a specific in-context example from the codebase.

Canadian Robotics Competition (CRC) Feb 2026 - Present
Software Developer & Problem Set Designer (Volunteer) / Montreal, QC
- Authored **12 algorithmic problems** (graph traversal, dynamic programming, greedy) for the upcoming CRC 2027 provincial programming competition, writing reference solutions and 20+ edge-case tests per problem in C++ and Python.

Self-Employed May 2025 - Present
Freelance Software Developer / Montreal, QC
- Delivered **12 projects for 9 clients** (web apps, automation, desktop GUIs), owning scoping, architecture, and deployment.
- Shipped a **cross-platform desktop application** in PySide6/Qt under a 6-day deadline, self-onboarding the framework from zero and delivering with no post-launch defects.
- Engineered a **Python automation pipeline** that cut a client's weekly workflow from about 8 hours to under 30 minutes (90% + faster).

PROJECTS

Facturo - Local-First Invoicing App Jun 2026 - Oct 2026
- Client project for a Québec trucking company (public demo, fictional data): built a **local-first invoicing app** that turns paper work tickets into tax-correct invoices (TPS/TVQ, Excel/PDF) and **reconciles supplier payments** by auto-matching statement rows to billed tickets and flagging duplicates.
- Added **AI extraction of handwritten tickets** with a local vision model (Ollama), snapping misreads to known values by edit distance and leaving uncertain fields blank rather than guessing.
- Engineered a **release pipeline** (GitHub Actions, PyInstaller) with SHA-256-verified in-app updates and automatic rollback, backed by **700+ tests** (pytest, Playwright e2e, Python/JS parity checks on discount math).

MAME Tool Rental - Tool & Machine Booking Platform Nov 2026 - Present
- Built a **full-stack rental platform** for the McGill Association of Mechanical Engineers (FastAPI, SQLite, Next.js, React): students book tools and machines with drawing uploads; admins review requests, track pickups and returns, and manage inventory from a stats dashboard.
- Shipped a **CI/CD pipeline** (GitHub Actions, Render, Vercel) gated by **1,300+ tests** (pytest, Vitest, Playwright), with per-visitor rate limiting and direct-to-API drawing uploads (up to 20 MB) past Vercel's proxy limit.

Atlantis - 2D Submarine Exploration Game 1st Place | Jan 2026
- **Architected 6 gameplay systems** and rebuilt save and inventory in the final 4 hours after a state-corruption bug.

CRC 2026 Competition Robot - Control Firmware 1st Place | Sep 2025 - Feb 2026
- Built the robot's **embedded state machine** in C/C++ as lead programmer, tuned **PID control loops**, and debugged hardware and software integration live between matches.